Online Resource 1 — Supplementary methods and results

Retinal foundation-model versus ImageNet pretraining for diabetic retinopathy grading across datasets: how much of the external performance drop reflects the reference standard?

Pravin Raj E (Independent Researcher, Chennai, Tamil Nadu, India; epravinraj0761@gmail.com; ORCID 0009-0000-2975-1631)

International Ophthalmology

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This file contains every analysis performed for the study that is not shown in full in the main article. Unless stated otherwise, intervals are 95% percentile bootstrap intervals (2,000 resamples, seed 42) and IDRiD denotes the **official** 516-image Disease-Grading release. Sections S10.1–S10.8 were run before the official IDRiD release was obtained and use a 455-image third-party mirror of IDRiD (audited in S1); their Messidor-2 and APTOS results are unaffected. Model abbreviations: baseline_efficientnet_b3 = EfficientNet-B3; baseline_resnet50 = ResNet-50; retfound_linprobe = RETFound linear probe; retfound_full = RETFound full fine-tuning. QWK, quadratic weighted kappa; ECE, expected calibration error; MSP, maximum softmax probability.

S1 Audit of the IDRiD mirror against the official release

The analysis was initially run on a widely shared third-party mirror of IDRiD. Comparison with the official IEEE DataPort release (doi:10.21227/H25W98) gave the following; all main-text IDRiD results use the official release.

- Official images with labels: 516 (train 413, test 103); grade counts 0-4: [168, 25, 168, 93, 62]; referable 62.6%
- Present in the mirror: 455; missing from mirror: 61
- Mirror IDs not in the official release: 0 []
- **Mirror labels that differ from the official grade: 27** (5.9% of mirror images)

id	official	mirror
IDRiD_098	3	4
IDRiD_099	3	4
IDRiD_192	3	4
IDRiD_194	1	0
IDRiD_198	1	0
IDRiD_201	2	0
IDRiD_208	2	0
IDRiD_215	2	0
IDRiD_256	1	0
IDRiD_260	2	1
IDRiD_263	2	0
IDRiD_266	1	0
IDRiD_270	2	0
IDRiD_278	3	4
IDRiD_282	2	1
IDRiD_290	1	0
IDRiD_299	3	4
IDRiD_311	2	1
IDRiD_324	3	4
IDRiD_349	3	2
IDRiD_412	2	1
IDRiD_062test	4	2
IDRiD_063test	1	0
IDRiD_064test	3	1
IDRiD_087test	4	2
IDRiD_088test	2	3
IDRiD_089test	2	3

S2 Duplicate and leakage checks

A byte-level (MD5) comparison of all 5,861 preprocessed images found **no** image shared between APTOS, Messidor-2 and IDRiD. Within APTOS, 41 of the 733 internal-test images have an exact duplicate in the training or validation split (a known property of APTOS 2019). Removing them changes internal QWK as follows (completed RETFound checkpoint): EfficientNet-B3 0.876 → 0.879; ResNet-50 0.875 → 0.880; RETFound linear probe 0.847 → 0.852; RETFound full 0.882 → 0.888 (n = 692). No internal comparison changes. A semantic near-duplicate check in an ImageNet feature space supported the byte-level result:

Feature extractor: ImageNet-pretrained EfficientNet-B3, 1536-d pooled, L2-normalised. Cosine similarity, all cross-dataset pairs.

Dataset pair	pairs cos>=0.90	>=0.95	>=0.98
APTOS <-> Messidor-2	47791	99	0
APTOS <-> IDRiD	303	0	0
Messidor-2 <-> IDRiD	507	0	0

APTOS internal-test images with a near-duplicate elsewhere

target pool	test images cos>=0.95	cos>=0.90
APTOS train/val	89 / 733	367 / 733
Messidor-2 (external)	7 / 733	214 / 733
IDRiD (external)	0 / 733	38 / 733

S3 Checkpoints, selection rule and the interrupted training run

Every scored checkpoint, the rule used to select it, and its SHA-256 hash. Model code and all training defaults were committed to version control before any external image was scored, and no external result was used for any modelling, checkpoint, threshold or calibration decision.

Model	Checkpoint	Selected by	SHA-256	Status
EfficientNet-B3 (primary baseline)	`outputs/phase4_baseline/efficientnet_b3_best.pt`	best APTOS-val QWK, patience 7	`a595133256a678cbf5ef6daca64842f989dc761c6f0bfe3d3b9926c480b`	**frozen**
ResNet-50 (secondary baseline)	`outputs/phase4_baseline/resnet50_best.pt`	best APTOS-val QWK, patience 7	`225610ffa51b778c6bfd3abd302aaad9cd68c7157131d9aa009a696520`	**frozen**
RETFound linear probe	`outputs/phase5_retfound/linprobe_best.pt`	best APTOS-val QWK, patience 7	`e7795a1bd2db209573e3a868cb927b30428db265a83b5e1ec081fe31c9`	**frozen**
RETFound full fine-tune (completed run, v4)	`outputs/phase5_retfound/full_best.pt` (copied from `C:\dr_ckpts_full_v4\full_best.pt`)	best APTOS-val QWK over the completed protocol (≤30 epochs, patience 7): epoch 21 of 28, val QWK 0.939	`b1d4a7b23b79413c2d313c4e742c79f7cf1dc6bfcf541868b511fd5933`	**frozen**
RETFound full fine-tune (superseded, provisional)	`outputs/phase5_retfound/full_best_provisional_epoch6.pt`	best APTOS-val QWK at epoch 6 of an interrupted run	`4344dd30356e7ae56143c0f6db670bc080db9b4d0[illegible]`	superseded; archived in `outputs/_archive_provisional_epoch6/`
Pretrained RETFound encoder	`weights/RETFound_mae_natureCFP.pth` (Hugging Face, gated)	—	`e1e4f66a1b792eeb6e2efaf158f33be35c8255f36[illegible]29da246485`	input weights

Interrupted run. The first fully fine-tuned RETFound run was stopped by hardware limits at epoch 6 of the planned 30 (validation QWK 0.935). Scored provisionally, it suggested a significantly larger external drop than EfficientNet-B3 (Messidor-2 QWK 0.368, IDRiD mirror 0.720). The pre-specified protocol was then completed (early stopping at epoch 28; best epoch 21; validation QWK 0.939), giving Messidor-2 QWK 0.487, and the conclusion changed from “worse” to “no detectable difference”. Both checkpoints are listed above; all results in the article use the completed run.

S4 Full performance on the official IDRiD release

model	dataset	n	qwk	qwk_lo	qwk_hi	accuracy	macro_f1	balanced_acc	auc_ovr
baseline_efficientnet_b3	APTOS-internal	733	0.876	0.849	0.901	0.804	0.644	0.641	0.928
baseline_efficientnet_b3	Messidor-2	1744	0.499	0.451	0.543	0.626	0.36	0.353	0.775
baseline_efficientnet_b3	IDRiD	516	0.726	0.681	0.767	0.465	0.416	0.451	0.804
baseline_resnet50	APTOS-internal	733	0.875	0.848	0.899	0.789	0.616	0.63	0.924
baseline_resnet50	Messidor-2	1744	0.435	0.381	0.483	0.614	0.354	0.339	0.754
baseline_resnet50	IDRiD	516	0.662	0.605	0.71	0.422	0.38	0.434	0.804
retfound_linprobe	APTOS-internal	733	0.847	0.819	0.874	0.759	0.607	0.633	0.908
retfound_linprobe	Messidor-2	1744	0.394	0.338	0.446	0.589	0.347	0.423	0.758
retfound_linprobe	IDRiD	516	0.59	0.534	0.642	0.312	0.302	0.381	0.762
retfound_full	APTOS-internal	733	0.882	0.858	0.906	0.783	0.632	0.647	0.9
retfound_full	Messidor-2	1744	0.487	0.435	0.535	0.624	0.368	0.35	0.75
retfound_full	IDRiD	516	0.703	0.655	0.747	0.403	0.371	0.453	0.793

S5 Referable diabetic retinopathy (grade ≥ 2)

1. Headline metrics with 95% CI

Model	Dataset	n	n-ref	Sens	Spec	AUROC	AUPRC	Sens@85%spec	Spec@85%sens
baseline_efficientnet_b3	APTOS-internal	733	298	0.903 (0.868-0.934)	0.931 (0.907-0.954)	0.980 (0.971-0.987)	0.968 (0.953-0.980)	0.987	0.947
baseline_efficientnet_b3	Messidor-2	1744	457	0.387 (0.345-0.433)	0.974 (0.966-0.983)	0.842 (0.819-0.864)	0.732 (0.694-0.771)	0.702	0.576
baseline_efficientnet_b3	IDRiD	516	323	0.827 (0.783-0.866)	0.907 (0.865-0.945)	0.927 (0.904-0.948)	0.961 (0.946-0.973)	0.854	0.855
baseline_resnet50	APTOS-internal	733	298	0.869 (0.829-0.905)	0.954 (0.935-0.973)	0.978 (0.968-0.986)	0.963 (0.944-0.979)	0.983	0.954
baseline_resnet50	Messidor-2	1744	457	0.357 (0.312-0.399)	0.956 (0.944-0.966)	0.789 (0.764-0.814)	0.630 (0.585-0.675)	0.589	0.487
baseline_resnet50	IDRiD	516	323	0.715 (0.667-0.764)	0.953 (0.920-0.980)	0.912 (0.887-0.936)	0.951 (0.933-0.966)	0.842	0.839
retfound_linprobe	APTOS-internal	733	298	0.859 (0.820-0.896)	0.952 (0.930-0.971)	0.976 (0.966-0.986)	0.969 (0.955-0.981)	0.977	0.956
retfound_linprobe	Messidor-2	1744	457	0.429 (0.383-0.473)	0.902 (0.885-0.918)	0.781 (0.754-0.806)	0.616 (0.571-0.661)	0.549	0.498
retfound_linprobe	IDRiD	516	323	0.827 (0.784-0.868)	0.658 (0.586-0.726)	0.882 (0.853-0.909)	0.936 (0.916-0.952)	0.759	0.710
retfound_full	APTOS-internal	733	298	0.909 (0.877-0.940)	0.952 (0.931-0.972)	0.967 (0.950-0.983)	0.968 (0.954-0.981)	0.973	0.968
retfound_full	Messidor-2	1744	457	0.392 (0.347-0.434)	0.975 (0.966-0.984)	0.817 (0.791-0.840)	0.716 (0.678-0.752)	0.659	0.521
retfound_full	IDRiD	516	323	0.734 (0.684-0.781)	0.927 (0.888-0.959)	0.876 (0.844-0.906)	0.942 (0.925-0.957)	0.814	0.642

2. Internal->external AUROC drop with 95% CI

Mirrors the primary-task QWK-drop table (Phase 9 sec. 2) for the referable-DR binarisation, so the manuscript can state whether the generalisation gap persists beyond 5-class ordinal grading.

Model	External set	AUROC internal	AUROC external	Drop (95% CI)
baseline_efficientnet_b3	Messidor-2	0.980	0.842	0.138 (0.115-0.161)
baseline_efficientnet_b3	IDRiD	0.980	0.927	0.053 (0.031-0.076)
baseline_resnet50	Messidor-2	0.978	0.789	0.189 (0.162-0.215)
baseline_resnet50	IDRiD	0.978	0.912	0.065 (0.040-0.094)
retfound_linprobe	Messidor-2	0.976	0.781	0.195 (0.169-0.223)
retfound_linprobe	IDRiD	0.976	0.882	0.094 (0.066-0.125)
retfound_full	Messidor-2	0.967	0.817	0.150 (0.122-0.180)
retfound_full	IDRiD	0.967	0.876	0.091 (0.056-0.126)

Paired DeLong comparison of referable-DR AUC versus EfficientNet-B3 (Holm adjustment over nine contrasts)

dataset	model	auc_model	auc_effb3	diff	lo	hi	p	p_holm
APTOS-internal	ResNet-50	0.9779	0.9799	-0.002	-0.0081	0.0042	0.5301	0.9471
APTOS-internal	RETFound linear probe	0.9764	0.9799	-0.0034	-0.0128	0.0059	0.4736	0.9471
APTOS-internal	RETFound full fine-tune	0.967	0.9799	-0.0129	-0.0277	0.002	0.0891	0.3565
Messidor-2	ResNet-50	0.7889	0.8416	-0.0527	-0.0727	-0.0328	0	0
Messidor-2	RETFound linear probe	0.7812	0.8416	-0.0605	-0.0855	-0.0354	0	0
Messidor-2	RETFound full fine-tune	0.8167	0.8416	-0.025	-0.0465	-0.0035	0.0229	0.1143
IDRiD	ResNet-50	0.9124	0.9268	-0.0144	-0.0317	0.003	0.1045	0.3565
IDRiD	RETFound linear probe	0.8823	0.9268	-0.0445	-0.0693	-0.0197	0.0004	0.0031
IDRiD	RETFound full fine-tune	0.8761	0.9268	-0.0507	-0.0811	-0.0203	0.0011	0.0064

S6 Shift decomposition (full)

Grade mix (%)

Dataset	0	1	2	3	4	referable
APTOS-internal	49.2	10.1	27.3	5.3	8.0	40.7
Messidor-2	58.3	15.5	19.9	4.3	2.0	26.2
IDRiD	32.6	4.8	32.6	18.0	12.0	62.6

A/C/D. QWK drop (internal − external) under each correction, 95% CI

Reweighted = external images importance-weighted to the APTOS-test grade mix (removes prevalence shift). 4-level = grades 0 and 1 merged (removes the mild-boundary reference-standard disagreement).

Model	External	raw	reweighted	4-level	4-level + reweighted
EfficientNet-B3	Messidor-2	0.377 [0.327, 0.431]	0.293 [0.237, 0.351]	0.200 [0.126, 0.276]	0.141 [0.052, 0.234]
EfficientNet-B3	IDRiD	0.150 [0.099, 0.203]	0.177 [0.117, 0.240]	0.070 [-0.002, 0.143]	0.043 [-0.034, 0.125]
ResNet-50	Messidor-2	0.440 [0.385, 0.496]	0.379 [0.309, 0.454]	0.248 [0.169, 0.325]	0.211 [0.115, 0.310]
ResNet-50	IDRiD	0.212 [0.156, 0.275]	0.234 [0.171, 0.306]	0.154 [0.076, 0.236]	0.136 [0.051, 0.225]
RETFound linear probe	Messidor-2	0.453 [0.392, 0.515]	0.296 [0.251, 0.348]	0.306 [0.228, 0.386]	0.123 [0.057, 0.192]
RETFound linear probe	IDRiD	0.257 [0.196, 0.320]	0.353 [0.285, 0.417]	0.118 [0.042, 0.192]	0.188 [0.103, 0.270]
RETFound full fine-tune	Messidor-2	0.395 [0.345, 0.453]	0.284 [0.239, 0.333]	0.202 [0.134, 0.278]	0.117 [0.044, 0.194]
RETFound full fine-tune	IDRiD	0.180 [0.128, 0.233]	0.223 [0.164, 0.285]	0.115 [0.041, 0.190]	0.115 [0.038, 0.190]

Model	External	raw	reweighted	4-level	4-level + reweighted

Share of the raw QWK drop removed by each correction

Model	External	by reweighting	by 4-level	by both	residual drop
EfficientNet-B3	Messidor-2	22%	47%	63%	0.141
EfficientNet-B3	IDRiD	-18%	53%	71%	0.043
ResNet-50	Messidor-2	14%	44%	52%	0.211
ResNet-50	IDRiD	-10%	27%	36%	0.136
RETFound linear probe	Messidor-2	35%	32%	73%	0.123
RETFound linear probe	IDRiD	-37%	54%	27%	0.188
RETFound full fine-tune	Messidor-2	28%	49%	70%	0.117
RETFound full fine-tune	IDRiD	-24%	36%	36%	0.115

Does the primary comparison survive the corrections? Difference in drop vs EfficientNet-B3

Model	External	raw	reweighted	4-level + reweighted
ResNet-50	Messidor-2	+0.063 [+0.018, +0.107]	+0.086 [+0.017, +0.155]	+0.070 [-0.024, +0.163]
ResNet-50	IDRiD	+0.063 [+0.011, +0.116]	+0.057 [-0.003, +0.121]	+0.092 [+0.014, +0.172]
RETFound linear probe	Messidor-2	+0.076 [+0.020, +0.131]	+0.004 [-0.060, +0.069]	-0.019 [-0.106, +0.069]
RETFound linear probe	IDRiD	+0.108 [+0.043, +0.172]	+0.177 [+0.101, +0.249]	+0.144 [+0.053, +0.229]
RETFound full fine-tune	Messidor-2	+0.018 [-0.022, +0.059]	-0.009 [-0.059, +0.040]	-0.024 [-0.096, +0.042]
RETFound full fine-tune	IDRiD	+0.030 [-0.028, +0.086]	+0.046 [-0.023, +0.114]	+0.071 [-0.009, +0.150]

B. Prevalence-invariant performance

Per-grade recall

Model	Dataset	0	1	2	3	4
EfficientNet-B3	APTOS-internal	0.96	0.53	0.79	0.49	0.44
EfficientNet-B3	Messidor-2	0.97	0.01	0.22	0.13	0.43
EfficientNet-B3	IDRiD	0.42	0.60	0.65	0.28	0.31
ResNet-50	APTOS-internal	0.96	0.66	0.73	0.51	0.29
ResNet-50	Messidor-2	0.95	0.01	0.20	0.28	0.26
ResNet-50	IDRiD	0.43	0.72	0.49	0.42	0.11
RETFound linear probe	APTOS-internal	0.95	0.61	0.58	0.51	0.51
RETFound linear probe	Messidor-2	0.90	0.00	0.18	0.41	0.63
RETFound linear probe	IDRiD	0.17	0.48	0.20	0.70	0.35
RETFound full fine-tune	APTOS-internal	0.95	0.59	0.69	0.51	0.49
RETFound full fine-tune	Messidor-2	0.97	0.00	0.22	0.16	0.40
RETFound full fine-tune	IDRiD	0.39	0.76	0.32	0.67	0.13

Referable DR: sensitivity / specificity (prevalence-invariant), and PPV at own vs APTOS prevalence

Model	Dataset	prev.	sens [CI]	spec [CI]	PPV own	PPV @APTOS prev
EfficientNet-B3	APTOS-internal	0.41	0.90 [0.86, 0.93]	0.93 [0.90, 0.95]	0.90	0.90
EfficientNet-B3	Messidor-2	0.26	0.39 [0.34, 0.43]	0.97 [0.96, 0.98]	0.84	0.91
EfficientNet-B3	IDRiD	0.63	0.83 [0.78, 0.86]	0.91 [0.86, 0.94]	0.94	0.86
ResNet-50	APTOS-internal	0.41	0.87 [0.83, 0.90]	0.95 [0.93, 0.97]	0.93	0.93
ResNet-50	Messidor-2	0.26	0.36 [0.31, 0.40]	0.96 [0.94, 0.97]	0.74	0.85
ResNet-50	IDRiD	0.63	0.72 [0.66, 0.76]	0.95 [0.91, 0.98]	0.96	0.91
RETFound linear probe	APTOS-internal	0.41	0.86 [0.81, 0.89]	0.95 [0.93, 0.97]	0.92	0.92
RETFound linear probe	Messidor-2	0.26	0.43 [0.38, 0.47]	0.90 [0.88, 0.92]	0.61	0.75
RETFound linear probe	IDRiD	0.63	0.83 [0.78, 0.86]	0.66 [0.59, 0.72]	0.80	0.62
RETFound full fine-tune	APTOS-internal	0.41	0.91 [0.87, 0.94]	0.95 [0.93, 0.97]	0.93	0.93
RETFound full fine-tune	Messidor-2	0.26	0.39 [0.35, 0.44]	0.98 [0.97, 0.98]	0.85	0.92
RETFound full fine-tune	IDRiD	0.63	0.73 [0.68, 0.78]	0.93 [0.88, 0.96]	0.94	0.87

Where reference-standard grade-1 (mild) images are predicted

Model	Dataset	n	→0	→1	→2	→3	→4
EfficientNet-B3	APTOS-internal	74	8	39	26	0	1
EfficientNet-B3	Messidor-2	270	259	4	6	0	1
EfficientNet-B3	IDRiD	25	7	15	3	0	0
ResNet-50	APTOS-internal	74	8	49	17	0	0
ResNet-50	Messidor-2	270	255	2	13	0	0
ResNet-50	IDRiD	25	5	18	2	0	0
RETFound linear probe	APTOS-internal	74	10	45	14	1	4
RETFound linear probe	Messidor-2	270	242	0	17	0	11
RETFound linear probe	IDRiD	25	2	12	5	6	0
RETFound full fine-tune	APTOS-internal	74	10	44	16	3	1
RETFound full fine-tune	Messidor-2	270	266	0	2	1	1
RETFound full fine-tune	IDRiD	25	6	19	0	0	0

S7 Calibration and the label-smoothing control

EfficientNet-B3 retrained with label smoothing 0.1 (all other settings unchanged) is shown alongside the primary models.

ECE: 15 equal-width bins on max-probability, 95% bootstrap CI (B=500). Error AUROC: grade error.

Model	Dataset	QWK	ECE [CI]	Brier	mean entropy	mean max-prob	AUROC(err) entropy	AUROC(err) MSP
EfficientNet-B3	APTOS-internal	0.876	0.101 [0.081, 0.126]	0.300	0.266	0.903	0.841	0.840
EfficientNet-B3	Messidor-2	0.499	0.279 [0.259, 0.304]	0.644	0.277	0.905	0.681	0.682
EfficientNet-B3	IDRiD	0.726	0.320 [0.277, 0.362]	0.818	0.554	0.785	0.585	0.585
EfficientNet-B3 + label smoothing 0.1	APTOS-internal	0.867	0.196 [0.177, 0.229]	0.381	1.034	0.630	0.502	0.655
EfficientNet-B3 + label smoothing 0.1	Messidor-2	0.464	0.041 [0.029, 0.070]	0.577	1.210	0.577	0.639	0.639
EfficientNet-B3 + label smoothing 0.1	IDRiD	0.681	0.185 [0.151, 0.232]	0.778	1.081	0.579	0.452	0.467
ResNet-50	APTOS-internal	0.875	0.058 [0.046, 0.091]	0.311	0.459	0.826	0.788	0.800
ResNet-50	Messidor-2	0.435	0.264 [0.247, 0.288]	0.650	0.362	0.878	0.652	0.654
ResNet-50	IDRiD	0.662	0.282 [0.246, 0.331]	0.837	0.760	0.704	0.500	0.499
RETFound linear probe	APTOS-internal	0.847	0.182 [0.159, 0.209]	0.395	1.108	0.583	0.695	0.761

Model	Dataset	QWK	ECE [CI]	Brier	mean entropy	mean max-prob	AUROC(err) entropy	AUROC(err) MSP
RETFound linear probe	Messidor-2	0.394	0.121 [0.102, 0.145]	0.594	1.355	0.468	0.649	0.655
RETFound linear probe	IDRiD	0.590	0.154 [0.115, 0.191]	0.794	1.270	0.461	0.600	0.579
RETFound full fine-tune	APTOS-internal	0.882	0.179 [0.165, 0.211]	0.369	0.972	0.656	0.516	0.635
RETFound full fine-tune	Messidor-2	0.487	0.095 [0.076, 0.115]	0.563	1.285	0.530	0.633	0.653
RETFound full fine-tune	IDRiD	0.703	0.170 [0.135, 0.215]	0.762	1.104	0.552	0.503	0.518

Difference in internal-to-external QWK drop (paired bootstrap, B=2000)

External	Comparison	ΔDrop	95% CI	p
Messidor-2	EffB3+LS minus EfficientNet-B3	+0.026	[-0.014, +0.067]	0.176
Messidor-2	EffB3+LS minus RETFound full fine-tune	+0.008	[-0.036, +0.054]	0.702
IDRiD	EffB3+LS minus EfficientNet-B3	+0.037	[-0.004, +0.079]	0.070
IDRiD	EffB3+LS minus RETFound full fine-tune	+0.007	[-0.053, +0.065]	0.825

S8 Uncertainty, error detection and selective referral (full)

Scores: msp = 1 − maximum softmax probability; margin = −(top-1 − top-2 probability); entropy = predictive entropy; grade SD = standard deviation of the predicted grade distribution; random = fixed random ordering. Grade SD is confounded by the predicted grade and is not recommended (it falls below chance for RETFound).

Scores rank cases from most to least trustworthy; the least trustworthy are referred. MSP = 1 - max softmax probability (simple confidence baseline). Paired bootstrap differences use identical resamples.

1. Error detection (grade error), AUROC [95% CI]

Model	Dataset	err. rate	MSP	margin	entropy	entropy T=20	grade SD
EfficientNet-B3	APTOS-internal	0.20	0.840 [0.81, 0.87]	0.839 [0.81, 0.87]	0.841 [0.81, 0.87]	—	0.820 [0.79, 0.85]
EfficientNet-B3	Messidor-2	0.37	0.682 [0.65, 0.71]	0.682 [0.65, 0.71]	0.681 [0.65, 0.71]	—	0.677 [0.65, 0.70]
EfficientNet-B3	IDRiD	0.53	0.585 [0.54, 0.63]	0.578 [0.53, 0.63]	0.585 [0.54, 0.64]	—	0.539 [0.49, 0.59]
ResNet-50	APTOS-internal	0.21	0.800 [0.77, 0.83]	0.806 [0.77, 0.84]	0.788 [0.75, 0.82]	—	0.756 [0.72, 0.79]
ResNet-50	Messidor-2	0.39	0.654 [0.63, 0.68]	0.655 [0.63, 0.68]	0.652 [0.63, 0.68]	—	0.645 [0.62, 0.67]
ResNet-50	IDRiD	0.58	0.499 [0.45, 0.55]	0.500 [0.45, 0.55]	0.500 [0.45, 0.55]	—	0.480 [0.43, 0.53]
RETFound linear probe	APTOS-internal	0.24	0.761 [0.72, 0.81]	0.773 [0.73, 0.82]	0.695 [0.65, 0.74]	—	0.332 [0.29, 0.38]
RETFound linear probe	Messidor-2	0.41	0.655 [0.63, 0.68]	0.651 [0.62, 0.68]	0.649 [0.62, 0.67]	—	0.499 [0.47, 0.53]
RETFound linear probe	IDRiD	0.69	0.579 [0.52, 0.63]	0.541 [0.49, 0.59]	0.600 [0.54, 0.65]	—	0.580 [0.52, 0.64]
RETFound full fine-tune	APTOS-internal	0.22	0.635 [0.57, 0.69]	0.666 [0.61, 0.72]	0.516 [0.46, 0.58]	—	0.263 [0.23, 0.30]
RETFound full fine-tune	Messidor-2	0.38	0.653 [0.63, 0.68]	0.659 [0.63, 0.69]	0.633 [0.61, 0.66]	—	0.418 [0.39, 0.45]
RETFound full fine-tune	IDRiD	0.60	0.518 [0.47, 0.57]	0.525 [0.47, 0.57]	0.503 [0.45, 0.55]	—	0.484 [0.43, 0.54]

2. Difference vs MSP baseline (grade error): ΔAUROC [95% CI]

Model	Dataset	margin	entropy	entropy T=20	grade SD
EfficientNet-B3	APTOS-internal	-0.001 [-0.004, +0.002]	+0.000 [-0.005, +0.005]	—	-0.021 [-0.033, -0.008]
EfficientNet-B3	Messidor-2	+0.000 [-0.001, +0.001]	-0.000 [-0.002, +0.001]	—	-0.004 [-0.009, +0.000]
EfficientNet-B3	IDRiD	-0.006 [-0.013, -0.000]	+0.001 [-0.012, +0.014]	—	-0.046 [-0.072, -0.019]
ResNet-50	APTOS-internal	+0.006 [+0.001, +0.011]	-0.011 [-0.020, -0.002]	—	-0.044 [-0.061, -0.027]
ResNet-50	Messidor-2	+0.001 [+0.000, +0.003]	-0.001 [-0.003, +0.000]	—	-0.009 [-0.015, -0.003]
ResNet-50	IDRiD	+0.001 [-0.009, +0.010]	+0.000 [-0.015, +0.017]	—	-0.020 [-0.047, +0.009]
RETFound linear probe	APTOS-internal	+0.012 [+0.002, +0.022]	-0.067 [-0.089, -0.045]	—	-0.429 [-0.481, -0.374]
RETFound linear probe	Messidor-2	-0.004 [-0.011, +0.002]	-0.006 [-0.015, +0.002]	—	-0.156 [-0.188, -0.125]
RETFound linear probe	IDRiD	-0.038 [-0.062, -0.011]	+0.021 [-0.003, +0.045]	—	+0.001 [-0.043, +0.043]
RETFound full fine-tune	APTOS-internal	+0.031 [+0.013, +0.051]	-0.118 [-0.153, -0.088]	—	-0.372 [-0.425, -0.317]
RETFound full fine-tune	Messidor-2	+0.006 [+0.000, +0.012]	-0.020 [-0.029, -0.011]	—	-0.234 [-0.278, -0.193]
RETFound full fine-tune	IDRiD	+0.006 [-0.011, +0.023]	-0.015 [-0.038, +0.007]	—	-0.034 [-0.088, +0.017]

3. Error detection (referable-DR error), AUROC [95% CI]

Model	Dataset	err. rate	MSP	entropy	grade SD
EfficientNet-B3	APTOS-internal	0.08	0.828 [0.79, 0.87]	0.822 [0.78, 0.86]	0.796 [0.75, 0.84]
EfficientNet-B3	Messidor-2	0.18	0.724 [0.69, 0.76]	0.725 [0.69, 0.76]	0.728 [0.70, 0.76]
EfficientNet-B3	IDRiD	0.14	0.577 [0.51, 0.64]	0.578 [0.50, 0.65]	0.541 [0.47, 0.61]
ResNet-50	APTOS-internal	0.08	0.750 [0.70, 0.80]	0.738 [0.69, 0.78]	0.732 [0.68, 0.78]
ResNet-50	Messidor-2	0.20	0.702 [0.67, 0.73]	0.701 [0.67, 0.73]	0.700 [0.67, 0.73]
ResNet-50	IDRiD	0.20	0.484 [0.43, 0.55]	0.465 [0.41, 0.53]	0.487 [0.42, 0.55]
RETFound linear probe	APTOS-internal	0.09	0.810 [0.75, 0.86]	0.814 [0.76, 0.86]	0.496 [0.44, 0.55]
RETFound linear probe	Messidor-2	0.22	0.695 [0.67, 0.72]	0.698 [0.67, 0.73]	0.558 [0.52, 0.59]
RETFound linear probe	IDRiD	0.24	0.677 [0.62, 0.73]	0.747 [0.70, 0.79]	0.726 [0.68, 0.77]
RETFound full fine-tune	APTOS-internal	0.07	0.690 [0.58, 0.79]	0.646 [0.54, 0.74]	0.367 [0.32, 0.41]
RETFound full fine-tune	Messidor-2	0.18	0.749 [0.72, 0.78]	0.757 [0.73, 0.79]	0.449 [0.41, 0.48]
RETFound full fine-tune	IDRiD	0.19	0.548 [0.48, 0.61]	0.591 [0.53, 0.65]	0.622 [0.58, 0.67]

4. Selective prediction with the MSP baseline: accuracy / QWK on retained cases

Gain vs. no referral with 95% CI. Random referral has expected gain 0.

Model	Dataset	referral	acc (gain [CI])	QWK (gain [CI])	errors captured	ref. sens / spec
EfficientNet-B3	APTOS-internal	0%	0.804	0.876	—	0.90 / 0.93
		10%	0.847 (+0.043 [+0.031, +0.053])	0.901 (+0.025 [+0.010, +0.040])	0.30	0.92 / 0.95
		20%	0.881 (+0.077 [+0.061, +0.097])	0.925 (+0.049 [+0.032, +0.072])	0.51	0.95 / 0.96
		30%	0.916 (+0.113 [+0.089, +0.135])	0.942 (+0.066 [+0.043, +0.090])	0.70	0.97 / 0.98
EfficientNet-B3	Messidor-2	0%	0.626	0.499	—	0.39 / 0.97
		10%	0.653 (+0.027 [+0.019, +0.035])	0.442 (-0.057 [-0.101, -0.023])	0.17	0.31 / 0.99
		20%	0.683 (+0.058 [+0.046, +0.069])	0.334 (-0.165 [-0.244, -0.095])	0.32	0.19 / 1.00
		30%	0.714 (+0.089 [+0.073, +0.103])	0.188 (-0.311 [-0.424, -0.208])	0.47	0.06 / 1.00

Model	Dataset	referral	acc (gain [CI])	QWK (gain [CI])	errors captured	ref. sens / spec
EfficientNet-B3	IDRiD	0%	0.465	0.726	—	0.83 / 0.91
		10%	0.476 (+0.011 [-0.002, +0.026])	0.741 (+0.014 [-0.011, +0.044])	0.12	0.82 / 0.92
		20%	0.487 (+0.022 [+0.001, +0.041])	0.744 (+0.018 [-0.012, +0.049])	0.23	0.81 / 0.95
		30%	0.501 (+0.036 [+0.010, +0.061])	0.742 (+0.016 [-0.018, +0.057])	0.35	0.81 / 0.96
ResNet-50	APTOS-internal	0%	0.789	0.875	—	0.87 / 0.95
		10%	0.817 (+0.028 [+0.017, +0.042])	0.886 (+0.012 [+0.003, +0.023])	0.22	0.86 / 0.96
		20%	0.852 (+0.063 [+0.044, +0.079])	0.899 (+0.024 [+0.007, +0.039])	0.44	0.86 / 0.98
		30%	0.885 (+0.096 [+0.071, +0.118])	0.913 (+0.039 [+0.013, +0.059])	0.62	0.84 / 0.99
ResNet-50	Messidor-2	0%	0.614	0.435	—	0.36 / 0.96
		10%	0.639 (+0.025 [+0.017, +0.033])	0.380 (-0.054 [-0.096, -0.017])	0.16	0.27 / 0.98
		20%	0.660 (+0.047 [+0.036, +0.059])	0.218 (-0.216 [-0.289, -0.156])	0.30	0.12 / 1.00
		30%	0.681 (+0.068 [+0.053, +0.085])	0.072 (-0.363 [-0.449, -0.293])	0.42	0.03 / 1.00
ResNet-50	IDRiD	0%	0.422	0.662	—	0.72 / 0.95
		10%	0.425 (+0.002 [-0.012, +0.017])	0.684 (+0.022 [-0.003, +0.058])	0.10	0.71 / 0.96
		20%	0.426 (+0.004 [-0.015, +0.025])	0.694 (+0.031 [-0.000, +0.068])	0.20	0.69 / 0.95
		30%	0.421 (-0.001 [-0.028, +0.027])	0.701 (+0.039 [+0.001, +0.086])	0.30	0.66 / 0.97
RETFound linear probe	APTOS-internal	0%	0.759	0.847	—	0.86 / 0.95
		10%	0.798 (+0.040 [+0.028, +0.050])	0.878 (+0.031 [+0.018, +0.046])	0.25	0.88 / 0.97
		20%	0.838 (+0.079 [+0.058, +0.094])	0.904 (+0.057 [+0.033, +0.073])	0.46	0.89 / 0.99
		30%	0.867 (+0.109 [+0.086, +0.131])	0.923 (+0.076 [+0.052, +0.097])	0.62	0.90 / 0.99
RETFound linear probe	Messidor-2	0%	0.589	0.394	—	0.43 / 0.90
		10%	0.612 (+0.023 [+0.015, +0.030])	0.415 (+0.022 [-0.005, +0.046])	0.15	0.42 / 0.92
		20%	0.634 (+0.044 [+0.033, +0.056])	0.438 (+0.044 [+0.004, +0.083])	0.29	0.40 / 0.94
		30%	0.648 (+0.058 [+0.044, +0.074])	0.450 (+0.056 [+0.007, +0.110])	0.40	0.35 / 0.95
RETFound linear probe	IDRiD	0%	0.312	0.590	—	0.83 / 0.66
		10%	0.317 (+0.005 [-0.009, +0.019])	0.619 (+0.029 [+0.005, +0.054])	0.11	0.84 / 0.68
		20%	0.327 (+0.015 [-0.004, +0.035])	0.661 (+0.071 [+0.040, +0.108])	0.22	0.83 / 0.73
		30%	0.330 (+0.018 [-0.005, +0.046])	0.700 (+0.110 [+0.068, +0.155])	0.32	0.84 / 0.78
RETFound full fine-tune	APTOS-internal	0%	0.783	0.882	—	0.91 / 0.95
		10%	0.821 (+0.038 [+0.026, +0.049])	0.913 (+0.031 [+0.017, +0.044])	0.26	0.92 / 0.97
		20%	0.852 (+0.068 [+0.051, +0.086])	0.924 (+0.041 [+0.025, +0.057])	0.45	0.94 / 0.98
		30%	0.867 (+0.084 [+0.063, +0.105])	0.928 (+0.046 [+0.029, +0.065])	0.57	0.94 / 0.99
RETFound full fine-tune	Messidor-2	0%	0.624	0.487	—	0.39 / 0.98
		10%	0.653 (+0.028 [+0.021, +0.036])	0.535 (+0.047 [+0.020, +0.066])	0.17	0.40 / 0.99
		20%	0.672 (+0.048 [+0.037, +0.059])	0.520 (+0.032 [-0.010, +0.070])	0.30	0.37 / 1.00

Model	Dataset	referral	acc (gain [CI])	QWK (gain [CI])	errors captured	ref. sens / spec
RETFound full fine-tune	IDRiD	30%	0.699 (+0.074 [+0.060, +0.088])	0.556 (+0.069 [+0.013, +0.110])	0.44	0.40 / 1.00
		0%	0.403	0.703	—	0.73 / 0.93
		10%	0.409 (+0.006 [-0.007, +0.020])	0.732 (+0.029 [+0.010, +0.054])	0.11	0.75 / 0.95
		20%	0.407 (+0.004 [-0.017, +0.026])	0.748 (+0.046 [+0.015, +0.078])	0.20	0.76 / 0.96
		30%	0.416 (+0.012 [-0.016, +0.040])	0.753 (+0.050 [+0.013, +0.088])	0.31	0.76 / 0.97

4b. What gets referred? Human-in-the-loop view (MSP ranking)

Referred cases go to a human grader. `missed referable` = referable-DR cases that are kept AND called non-referable, as a fraction of all referable cases (the system's miss rate if the grader is correct on referred cases).

Model	Dataset	referral	ref. prev. kept	ref. prev. referred	missed referable
EfficientNet-B3	APTOS-internal	0%	0.41	nan	0.10
		10%	0.37	0.75	0.07
		20%	0.33	0.71	0.03
		30%	0.28	0.70	0.02
EfficientNet-B3	Messidor-2	0%	0.26	nan	0.61
		10%	0.21	0.71	0.50
		20%	0.16	0.68	0.39
		30%	0.11	0.61	0.28
EfficientNet-B3	IDRiD	0%	0.63	nan	0.17
		10%	0.63	0.62	0.16
		20%	0.62	0.66	0.15
		30%	0.61	0.67	0.13
ResNet-50	APTOS-internal	0%	0.41	nan	0.13
		10%	0.36	0.84	0.11
		20%	0.31	0.80	0.09
		30%	0.24	0.80	0.06
ResNet-50	Messidor-2	0%	0.26	nan	0.64
		10%	0.22	0.63	0.56
		20%	0.17	0.61	0.47
		30%	0.14	0.55	0.36
ResNet-50	IDRiD	0%	0.63	nan	0.28
		10%	0.60	0.87	0.25
		20%	0.58	0.83	0.23
		30%	0.55	0.81	0.21
RETFound linear probe	APTOS-internal	0%	0.41	nan	0.14
		10%	0.36	0.82	0.10
		20%	0.31	0.78	0.07
		30%	0.25	0.78	0.04
RETFound linear probe	Messidor-2	0%	0.26	nan	0.57
		10%	0.24	0.49	0.47
		20%	0.21	0.45	0.39
		30%	0.19	0.42	0.33
RETFound linear probe	IDRiD	0%	0.63	nan	0.17
		10%	0.64	0.50	0.15
		20%	0.66	0.50	0.14
		30%	0.67	0.53	0.12
RETFound full fine-tune	APTOS-internal	0%	0.41	nan	0.09
		10%	0.37	0.73	0.07
		20%	0.34	0.68	0.04
		30%	0.31	0.63	0.03

Model	Dataset	referral	ref. prev. kept	ref. prev. referred	missed referable
RETFound full fine-tune	Messidor-2	0%	0.26	nan	0.61
		10%	0.23	0.57	0.47
		20%	0.19	0.54	0.37
		30%	0.16	0.49	0.26
RETFound full fine-tune	IDRiD	0%	0.63	nan	0.27
		10%	0.65	0.44	0.23
		20%	0.67	0.44	0.21
		30%	0.70	0.45	0.19

5. Does entropy beat MSP for referral? Accuracy at the same referral rate

Model	Dataset	referral	MSP acc	entropy acc	Δ [95% CI]
EfficientNet-B3	APTOS-internal	10%	0.847	0.847	+0.000 [-0.009, +0.011]
EfficientNet-B3	APTOS-internal	20%	0.881	0.889	+0.009 [-0.007, +0.012]
EfficientNet-B3	APTOS-internal	30%	0.916	0.920	+0.004 [-0.008, +0.006]
EfficientNet-B3	Messidor-2	10%	0.653	0.654	+0.001 [-0.004, +0.005]
EfficientNet-B3	Messidor-2	20%	0.683	0.684	+0.001 [-0.002, +0.005]
EfficientNet-B3	Messidor-2	30%	0.714	0.712	-0.002 [-0.005, +0.002]
EfficientNet-B3	IDRiD	10%	0.476	0.481	+0.004 [-0.009, +0.019]
EfficientNet-B3	IDRiD	20%	0.487	0.489	+0.002 [-0.015, +0.022]
EfficientNet-B3	IDRiD	30%	0.501	0.501	+0.000 [-0.019, +0.022]
ResNet-50	APTOS-internal	10%	0.817	0.814	-0.003 [-0.014, +0.005]
ResNet-50	APTOS-internal	20%	0.852	0.846	-0.005 [-0.017, +0.012]
ResNet-50	APTOS-internal	30%	0.885	0.879	-0.006 [-0.019, +0.010]
ResNet-50	Messidor-2	10%	0.639	0.636	-0.003 [-0.007, +0.003]
ResNet-50	Messidor-2	20%	0.660	0.662	+0.002 [-0.005, +0.006]
ResNet-50	Messidor-2	30%	0.681	0.682	+0.001 [-0.003, +0.004]
ResNet-50	IDRiD	10%	0.425	0.429	+0.004 [-0.013, +0.013]
ResNet-50	IDRiD	20%	0.426	0.433	+0.007 [-0.010, +0.024]
ResNet-50	IDRiD	30%	0.421	0.429	+0.008 [-0.017, +0.033]
RETFound linear probe	APTOS-internal	10%	0.798	0.794	-0.005 [-0.017, +0.006]
RETFound linear probe	APTOS-internal	20%	0.838	0.809	-0.029 [-0.039, -0.009]
RETFound linear probe	APTOS-internal	30%	0.867	0.840	-0.027 [-0.045, -0.010]
RETFound linear probe	Messidor-2	10%	0.612	0.612	+0.000 [-0.008, +0.006]
RETFound linear probe	Messidor-2	20%	0.634	0.628	-0.006 [-0.016, +0.002]
RETFound linear probe	Messidor-2	30%	0.648	0.649	+0.002 [-0.010, +0.012]
RETFound linear probe	IDRiD	10%	0.317	0.317	+0.000 [-0.013, +0.013]
RETFound linear probe	IDRiD	20%	0.327	0.332	+0.005 [-0.015, +0.019]
RETFound linear probe	IDRiD	30%	0.330	0.343	+0.014 [-0.011, +0.033]
RETFound full fine-tune	APTOS-internal	10%	0.821	0.808	-0.014 [-0.026, -0.003]
RETFound full fine-tune	APTOS-internal	20%	0.852	0.824	-0.027 [-0.044, -0.012]
RETFound full fine-tune	APTOS-internal	30%	0.867	0.819	-0.049 [-0.066, -0.029]
RETFound full fine-tune	Messidor-2	10%	0.653	0.652	-0.001 [-0.005, +0.003]
RETFound full fine-tune	Messidor-2	20%	0.672	0.670	-0.003 [-0.009, +0.004]
RETFound full fine-tune	Messidor-2	30%	0.699	0.697	-0.002 [-0.010, +0.005]
RETFound full fine-tune	IDRiD	10%	0.409	0.407	-0.002 [-0.017, +0.009]
RETFound full fine-tune	IDRiD	20%	0.407	0.400	-0.007 [-0.029, +0.015]
RETFound full fine-tune	IDRiD	30%	0.416	0.421	+0.006 [-0.022, +0.022]

6. AURC (grade error; lower = better), E-AURC = excess over an oracle ranking

Model	Dataset	risk@100%	MSP	margin	entropy	entropy T=20	grade SD	random
EfficientNet-B3	APTOS-internal	0.196	0.056	0.056	0.056	—	0.062	0.204
EfficientNet-B3	Messidor-2	0.374	0.252	0.252	0.252	—	0.253	0.375
EfficientNet-B3	IDRiD	0.535	0.454	0.457	0.455	—	0.488	0.508
ResNet-50	APTOS-internal	0.211	0.073	0.071	0.076	—	0.086	0.222
ResNet-50	Messidor-2	0.386	0.277	0.276	0.278	—	0.282	0.390
ResNet-50	IDRiD	0.578	0.558	0.556	0.562	—	0.596	0.590
RETFound linear probe	APTOS-internal	0.241	0.137	0.128	0.165	—	0.323	0.261
RETFound linear probe	Messidor-2	0.411	0.296	0.299	0.305	—	0.408	0.396
RETFound linear probe	IDRiD	0.688	0.631	0.662	0.612	—	0.613	0.729
RETFound full fine-tune	APTOS-internal	0.217	0.186	0.175	0.228	—	0.306	0.226
RETFound full fine-tune	Messidor-2	0.376	0.272	0.269	0.291	—	0.439	0.374
RETFound full fine-tune	IDRiD	0.597	0.585	0.584	0.589	—	0.562	0.610

S9 Saliency localisation against expert lesion masks (full)

IDRiD Segmentation subset, 81 images. Lift = share of saliency mass on lesions divided by lesion share of the field of view. Cross-image = the model's map for a different image (derangement, seed 42).

Lift = share of saliency mass on lesions / share of the field of view that is lesion (uniform map = 1). Cross-image = the model's map for a different image.

Map	energy on lesions	lift (any)	pointing hit	pixel AUROC	top-10% precision	lift MA	lift HE	lift EX	lift SE	energy on OD
Centre-Gaussian control	0.09 [0.07, 0.12]	1.28 [1.19, 1.38]	0.36 [0.25, 0.47]	0.59 [0.56, 0.61]	0.11 [0.08, 0.14]	1.23	0.90	1.80	0.70	0.03
EfficientNet-B3 (cross-image)	0.08 [0.06, 0.11]	1.17 [1.08, 1.25]	0.31 [0.21, 0.41]	0.55 [0.53, 0.58]	0.09 [0.06, 0.12]	1.18	0.91	1.49	0.70	0.03
ResNet-50 (cross-image)	0.08 [0.06, 0.11]	1.15 [1.08, 1.22]	0.33 [0.23, 0.44]	0.56 [0.53, 0.58]	0.09 [0.07, 0.12]	1.18	0.88	1.46	0.80	0.03
RETFound full fine-tune (cross-image)	0.07 [0.05, 0.10]	1.02 [0.94, 1.11]	0.25 [0.16, 0.35]	0.51 [0.49, 0.53]	0.08 [0.05, 0.11]	1.08	0.91	1.13	0.81	0.04
RETFound linear probe (cross-image)	0.08 [0.06, 0.11]	1.12 [1.04, 1.21]	0.36 [0.26, 0.47]	0.55 [0.52, 0.57]	0.09 [0.06, 0.12]	1.10	1.03	1.30	0.83	0.03
EfficientNet-B3	0.09 [0.07, 0.13]	1.37 [1.28, 1.48]	0.48 [0.37, 0.59]	0.61 [0.58, 0.63]	0.12 [0.09, 0.15]	1.21	1.03	1.73	0.87	0.03
ResNet-50	0.09 [0.07, 0.12]	1.32 [1.25, 1.41]	0.49 [0.38, 0.60]	0.61 [0.59, 0.64]	0.12 [0.09, 0.15]	1.21	1.02	1.61	0.95	0.03
RETFound full fine-tune	0.12 [0.09, 0.15]	1.79 [1.63, 1.95]	0.64 [0.53, 0.74]	0.67 [0.63, 0.70]	0.19 [0.15, 0.23]	1.24	1.46	1.83	2.24	0.02
RETFound linear probe	0.12 [0.09, 0.16]	1.67 [1.54, 1.82]	0.60 [0.51, 0.72]	0.67 [0.64, 0.70]	0.19 [0.15, 0.23]	1.20	1.46	1.74	1.77	0.02

Mean lesion share of the field of view: 0.071.

Paired comparisons (Wilcoxon signed-rank over images)

Model	vs	metric	model	comparator	Δ	images better	p
EfficientNet-B3	centre	lift_any	1.372	1.283	+0.090	0.57	0.099
EfficientNet-B3	centre	auroc_lesion	0.606	0.587	+0.020	0.55	0.082
EfficientNet-B3	centre	energy_any	0.094	0.090	+0.004	0.57	0.068
EfficientNet-B3	centre	top10_precision	0.121	0.107	+0.013	0.53	0.18
EfficientNet-B3	centre	energy_od	0.026	0.027	-0.001	0.42	0.62
EfficientNet-B3	cross_image	lift_any	1.372	1.165	+0.207	0.77	1.6e-08
EfficientNet-B3	cross_image	auroc_lesion	0.606	0.555	+0.051	0.75	2.3e-08
EfficientNet-B3	cross_image	energy_any	0.094	0.082	+0.012	0.77	1.2e-08
EfficientNet-B3	cross_image	top10_precision	0.121	0.091	+0.030	0.72	1.2e-06
EfficientNet-B3	cross_image	energy_od	0.026	0.032	-0.006	0.38	0.0088
ResNet-50	centre	lift_any	1.324	1.283	+0.041	0.54	0.45
ResNet-50	centre	auroc_lesion	0.612	0.587	+0.025	0.62	0.0078
ResNet-50	centre	energy_any	0.090	0.090	-0.001	0.54	0.75
ResNet-50	centre	top10_precision	0.119	0.107	+0.012	0.60	0.024
ResNet-50	centre	energy_od	0.032	0.027	+0.004	0.57	0.023
ResNet-50	cross_image	lift_any	1.324	1.147	+0.177	0.72	2.1e-07
ResNet-50	cross_image	auroc_lesion	0.612	0.555	+0.057	0.82	4.8e-10
ResNet-50	cross_image	energy_any	0.090	0.080	+0.010	0.72	6.9e-07
ResNet-50	cross_image	top10_precision	0.119	0.091	+0.028	0.56	0.0033
ResNet-50	cross_image	energy_od	0.032	0.031	+0.001	0.51	0.6
ResNet-50	EfficientNet-B3	lift_any	1.324	1.372	-0.049	0.42	0.17
ResNet-50	EfficientNet-B3	auroc_lesion	0.612	0.606	+0.006	0.51	0.23
ResNet-50	EfficientNet-B3	energy_any	0.090	0.094	-0.004	0.42	0.048
ResNet-50	EfficientNet-B3	top10_precision	0.119	0.121	-0.002	0.53	0.8
ResNet-50	EfficientNet-B3	energy_od	0.032	0.026	+0.005	0.77	1.4e-05
RETFound linear probe	centre	lift_any	1.672	1.283	+0.389	0.73	3e-06
RETFound linear probe	centre	auroc_lesion	0.670	0.587	+0.083	0.72	1.1e-05
RETFound linear probe	centre	energy_any	0.121	0.090	+0.031	0.73	1.1e-07
RETFound linear probe	centre	top10_precision	0.187	0.107	+0.079	0.72	8.9e-08
RETFound linear probe	centre	energy_od	0.016	0.027	-0.011	0.30	3.7e-06
RETFound linear probe	cross_image	lift_any	1.672	1.123	+0.548	0.79	2.9e-09
RETFound linear probe	cross_image	auroc_lesion	0.670	0.547	+0.123	0.80	2.2e-08
RETFound linear probe	cross_image	energy_any	0.121	0.079	+0.042	0.79	3.8e-09
RETFound linear probe	cross_image	top10_precision	0.187	0.089	+0.098	0.80	2.7e-09
RETFound linear probe	cross_image	energy_od	0.016	0.031	-0.015	0.30	1.8e-06
RETFound linear probe	EfficientNet-B3	lift_any	1.672	1.372	+0.299	0.72	0.00015
RETFound linear probe	EfficientNet-B3	auroc_lesion	0.670	0.606	+0.063	0.71	0.00022
RETFound linear probe	EfficientNet-B3	energy_any	0.121	0.094	+0.027	0.72	2.9e-06
RETFound linear probe	EfficientNet-B3	top10_precision	0.187	0.121	+0.066	0.73	2.2e-06
RETFound linear probe	EfficientNet-B3	energy_od	0.016	0.026	-0.010	0.33	4.9e-05
RETFound full fine-tune	centre	lift_any	1.788	1.283	+0.505	0.69	7e-06
RETFound full fine-tune	centre	auroc_lesion	0.668	0.587	+0.082	0.68	0.00014
RETFound full fine-tune	centre	energy_any	0.121	0.090	+0.031	0.69	1.1e-05
RETFound full fine-tune	centre	top10_precision	0.186	0.107	+0.079	0.72	3.3e-06
RETFound full fine-tune	centre	energy_od	0.019	0.027	-0.008	0.36	0.00073
RETFound full fine-tune	cross_image	lift_any	1.788	1.022	+0.766	0.85	1.1e-10
RETFound full fine-tune	cross_image	auroc_lesion	0.668	0.509	+0.159	0.82	9.1e-10
RETFound full fine-tune	cross_image	energy_any	0.121	0.073	+0.049	0.85	2.5e-11
RETFound full fine-tune	cross_image	top10_precision	0.186	0.078	+0.108	0.83	1.7e-11
RETFound full fine-tune	cross_image	energy_od	0.019	0.036	-0.017	0.26	3.3e-06
RETFound full fine-tune	EfficientNet-B3	lift_any	1.788	1.372	+0.416	0.65	0.00012
RETFound full fine-tune	EfficientNet-B3	auroc_lesion	0.668	0.606	+0.062	0.66	0.0008
RETFound full fine-tune	EfficientNet-B3	energy_any	0.121	0.094	+0.027	0.65	5.5e-05
RETFound full fine-tune	EfficientNet-B3	top10_precision	0.186	0.121	+0.065	0.69	2.8e-05
RETFound full fine-tune	EfficientNet-B3	energy_od	0.019	0.026	-0.007	0.40	0.0064

Resolution-matched control (Phase 26b)

All maps degraded to the CNN's native 7x7 grid (average-pool, bilinear back to 224).

Model	resolution	lift (any)	pixel AUROC	pointing hit	top-10% precision
EfficientNet-B3	7x7	1.32 [1.24, 1.41]	0.61 [0.58, 0.63]	0.47 [0.36, 0.58]	0.12 [0.09, 0.15]
EfficientNet-B3	native	1.37 [1.28, 1.48]	0.61 [0.58, 0.63]	0.48 [0.37, 0.59]	0.12 [0.09, 0.15]
ResNet-50	7x7	1.27 [1.20, 1.34]	0.61 [0.58, 0.63]	0.46 [0.36, 0.57]	0.12 [0.09, 0.15]
ResNet-50	native	1.32 [1.25, 1.41]	0.61 [0.59, 0.64]	0.49 [0.38, 0.60]	0.12 [0.09, 0.15]
RETFound full fine-tune	7x7	1.44 [1.34, 1.54]	0.66 [0.63, 0.69]	0.54 [0.43, 0.65]	0.16 [0.12, 0.20]
RETFound full fine-tune	native	1.79 [1.63, 1.95]	0.67 [0.63, 0.70]	0.64 [0.53, 0.74]	0.19 [0.15, 0.23]
RETFound linear probe	7x7	1.41 [1.33, 1.50]	0.65 [0.62, 0.68]	0.59 [0.49, 0.70]	0.16 [0.13, 0.20]
RETFound linear probe	native	1.67 [1.54, 1.82]	0.67 [0.64, 0.70]	0.60 [0.51, 0.72]	0.19 [0.15, 0.23]

RETFound (7x7) vs EfficientNet-B3 (native)	metric	RETFound	EffB3	Δ	images better	p
RETFound full fine-tune	lift_any	1.440	1.372	+0.067	0.58	0.17
RETFound full fine-tune	auroc_lesion	0.661	0.606	+0.055	0.68	0.00041
RETFound full fine-tune	top10_precision	0.157	0.121	+0.036	0.62	0.0029
RETFound linear probe	lift_any	1.409	1.372	+0.037	0.58	0.13
RETFound linear probe	auroc_lesion	0.650	0.606	+0.044	0.65	0.00095
RETFound linear probe	top10_precision	0.160	0.121	+0.039	0.65	0.00021

S10 Sensitivity analyses (IDRiD mirror)

S10.1 Primary inference on the IDRiD mirror

Generated 2026-10-21T06:16:34+00:00; B=2000, seed=42, preds=outputs/phase6

Confirmatory family (primary): difference in QWK drop vs EfficientNet-B3

Model	External	dDrop	95% CI	Bonferroni 98.75% CI	p (boot)	p (Holm)
RETFound full fine-tune	Messidor-2	+0.018	[-0.024, 0.058]	[-0.036, 0.068]	0.3938	0.7876
RETFound full fine-tune	IDRiD	+0.022	[-0.033, 0.083]	[-0.051, 0.096]	0.4548	0.7876
RETFound linear probe	Messidor-2	+0.076	[0.019, 0.132]	[0.004, 0.146]	0.0100	0.0300
RETFound linear probe	IDRiD	+0.105	[0.038, 0.170]	[0.021, 0.195]	0.0040	0.0160

Secondary: ResNet-50 vs EfficientNet-B3

External	dDrop	95% CI	p (boot)
Messidor-2	+0.063	[0.017, 0.109]	0.0050
IDRiD	+0.051	[0.003, 0.102]	0.0410

Paired QWK difference (model - EfficientNet-B3)

Model	Dataset	QWK model	QWK base	diff	95% CI
ResNet-50	APTOS-internal	0.875	0.876	-0.001	[-0.021, 0.019]
ResNet-50	Messidor-2	0.435	0.499	-0.064	[-0.107, -0.023]
ResNet-50	IDRiD	0.707	0.760	-0.052	[-0.098, -0.004]
RETFound linear probe	APTOS-internal	0.847	0.876	-0.029	[-0.054, -0.005]
RETFound linear probe	Messidor-2	0.394	0.499	-0.105	[-0.157, -0.053]
RETFound linear probe	IDRiD	0.626	0.760	-0.134	[-0.192, -0.075]
RETFound full fine-tune	APTOS-internal	0.882	0.876	+0.006	[-0.014, 0.029]
RETFound full fine-tune	Messidor-2	0.487	0.499	-0.011	[-0.047, 0.025]
RETFound full fine-tune	IDRiD	0.744	0.760	-0.016	[-0.068, 0.038]

Worst-case non-independence (m = n/2)

Model	External	dDrop	95% CI	excludes 0
RETFound full fine-tune	Messidor-2	+0.018	[-0.042, 0.078]	False
RETFound full fine-tune	IDRiD	+0.022	[-0.056, 0.105]	False
RETFound linear probe	Messidor-2	+0.076	[-0.004, 0.157]	False
RETFound linear probe	IDRiD	+0.105	[0.020, 0.199]	True
ResNet-50	Messidor-2	+0.063	[-0.002, 0.126]	False
ResNet-50	IDRiD	+0.051	[-0.020, 0.122]	False

IDRiD reweighted to the official 516-image grade mix

Model	QWK mirror	QWK reweighted	dDrop reweighted (95% CI)
EfficientNet-B3	0.760	0.765	—
ResNet-50	0.707	0.722	+0.042 [-0.006, 0.093]
RETFound linear probe	0.626	0.619	+0.117 [0.049, 0.184]
RETFound full fine-tune	0.744	0.750	+0.022 [-0.034, 0.081]

S10.2 Twenty-pass Monte Carlo dropout (stochastic layers active only where active in training)

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Confirmatory family (primary): difference in QWK drop vs EfficientNet-B3

Model	External	dDrop	95% CI	Bonferroni 98.75% CI	p (boot)	p (Holm)
RETFound full fine-tune	Messidor-2	+0.019	[-0.024, 0.062]	[-0.036, 0.073]	0.3988	0.4278
RETFound full fine-tune	IDRiD	+0.035	[-0.020, 0.092]	[-0.035, 0.107]	0.2139	0.4278
RETFound linear probe	Messidor-2	+0.071	[0.015, 0.126]	[0.002, 0.139]	0.0130	0.0390
RETFound linear probe	IDRiD	+0.105	[0.040, 0.169]	[0.020, 0.195]	0.0040	0.0160

Secondary: ResNet-50 vs EfficientNet-B3

External	dDrop	95% CI	p (boot)
Messidor-2	+0.057	[0.010, 0.108]	0.0210
IDRiD	+0.047	[-0.001, 0.096]	0.0570

Paired QWK difference (model - EfficientNet-B3)

Model	Dataset	QWK model	QWK base	diff	95% CI
ResNet-50	APTOS-internal	0.874	0.881	-0.007	[-0.028, 0.015]
ResNet-50	Messidor-2	0.435	0.499	-0.064	[-0.108, -0.022]
ResNet-50	IDRiD	0.710	0.764	-0.054	[-0.099, -0.007]
RETFound linear probe	APTOS-internal	0.847	0.881	-0.034	[-0.058, -0.008]
RETFound linear probe	Messidor-2	0.394	0.499	-0.105	[-0.156, -0.054]
RETFound linear probe	IDRiD	0.626	0.764	-0.139	[-0.197, -0.081]
RETFound full fine-tune	APTOS-internal	0.887	0.881	+0.006	[-0.013, 0.026]
RETFound full fine-tune	Messidor-2	0.486	0.499	-0.013	[-0.050, 0.027]
RETFound full fine-tune	IDRiD	0.735	0.764	-0.029	[-0.082, 0.022]

Worst-case non-independence (m = n/2)

Model	External	dDrop	95% CI	excludes 0
RETFound full fine-tune	Messidor-2	+0.019	[-0.040, 0.079]	False
RETFound full fine-tune	IDRiD	+0.035	[-0.043, 0.115]	False
RETFound linear probe	Messidor-2	+0.071	[-0.008, 0.151]	False
RETFound linear probe	IDRiD	+0.105	[0.019, 0.199]	True
ResNet-50	Messidor-2	+0.057	[-0.009, 0.126]	False
ResNet-50	IDRiD	+0.047	[-0.024, 0.117]	False

IDRiD reweighted to the official 516-image grade mix

Model	QWK mirror	QWK reweighted	dDrop reweighted (95% CI)
EfficientNet-B3	0.764	0.770	—
ResNet-50	0.710	0.725	+0.039 [-0.008, 0.086]
RETFound linear probe	0.626	0.619	+0.117 [0.051, 0.184]
RETFound full fine-tune	0.735	0.741	+0.035 [-0.022, 0.092]

S10.3 Training seeds

Extra full-fine-tune runs scored unmodified on the Phase-6 code path. Primary model = seed 42.

QWK across seeds (mean +/- SD [min, max])

Family	Dataset	seeds	QWK mean	SD	min	max	range
efficientnet_b3	APTOS-internal	4	0.879	0.003	0.876	0.883	0.007
efficientnet_b3	Messidor-2	4	0.500	0.024	0.469	0.527	0.058
efficientnet_b3	IDRiD	4	0.749	0.015	0.732	0.764	0.033
retfound_linprobe	APTOS-internal	4	0.855	0.008	0.847	0.865	0.018
retfound_linprobe	Messidor-2	4	0.395	0.015	0.374	0.408	0.034
retfound_linprobe	IDRiD	4	0.623	0.012	0.605	0.631	0.026

Internal->external QWK drop across seeds

Family	External	seeds	drop mean	SD	min	max
efficientnet_b3	Messidor-2	4	0.379	0.025	0.353	0.413
efficientnet_b3	IDRiD	4	0.130	0.015	0.116	0.148
retfound_linprobe	Messidor-2	4	0.461	0.017	0.444	0.484
retfound_linprobe	IDRiD	4	0.233	0.019	0.221	0.260

S10.4 Loss function and class weighting

QWK by family x loss x dataset

Family	Loss	Internal	Messidor-2	IDRiD
efficientnet_b3	weighted_ce	0.876	0.499	0.760
efficientnet_b3	ce_noweight	0.881	0.549	0.706
efficientnet_b3	focal_g2	0.877	0.543	0.769
retfound_linprobe	weighted_ce	0.847	0.394	0.626
retfound_linprobe	ce_noweight	0.851	0.316	0.751
retfound_linprobe	focal_g2	0.858	0.404	0.636

Internal->external QWK drop by family x loss

Family	Loss	Messidor-2 drop	IDRiD drop
efficientnet_b3	weighted_ce	0.377	0.116
efficientnet_b3	ce_noweight	0.332	0.175
efficientnet_b3	focal_g2	0.334	0.108
retfound_linprobe	weighted_ce	0.453	0.221
retfound_linprobe	ce_noweight	0.535	0.100
retfound_linprobe	focal_g2	0.454	0.222

S10.5 Inference-time input changes (contrast enhancement, colour transfer, resolution)

QWK by model x dataset x variant:

Model	Dataset	variant	QWK	acc	referable AUC
baseline_efficientnet_b3	APTOS-internal	baseline	0.876	0.803	0.980
baseline_efficientnet_b3	Messidor-2	baseline	0.499	0.626	0.842
baseline_efficientnet_b3	IDRiD	baseline	0.760	0.497	0.962
baseline_efficientnet_b3	APTOS-internal	clahe	0.807	0.641	0.977
baseline_efficientnet_b3	Messidor-2	clahe	0.458	0.606	0.792
baseline_efficientnet_b3	IDRiD	clahe	0.697	0.448	0.966
baseline_efficientnet_b3	APTOS-internal	colour_xfer	0.816	0.704	0.979
baseline_efficientnet_b3	Messidor-2	colour_xfer	0.512	0.608	0.809
baseline_efficientnet_b3	IDRiD	colour_xfer	0.689	0.464	0.960
retfound_full	APTOS-internal	baseline	0.882	0.783	0.967
retfound_full	Messidor-2	baseline	0.487	0.624	0.817
retfound_full	IDRiD	baseline	0.744	0.451	0.885
retfound_full	APTOS-internal	clahe	0.703	0.570	0.966
retfound_full	Messidor-2	clahe	0.096	0.588	0.733
retfound_full	IDRiD	clahe	0.410	0.380	0.938
retfound_full	APTOS-internal	colour_xfer	0.810	0.689	0.944
retfound_full	Messidor-2	colour_xfer	0.284	0.608	0.742
retfound_full	IDRiD	colour_xfer	0.693	0.505	0.936

Internal->external QWK drop, and change vs the baseline variant (negative = the variant *recovers* generalisation):

Model	External	variant	QWK drop	delta vs baseline
baseline_efficientnet_b3	Messidor-2	baseline	0.377	+0.000
baseline_efficientnet_b3	Messidor-2	clahe	0.350	-0.028
baseline_efficientnet_b3	Messidor-2	colour_xfer	0.304	-0.073
baseline_efficientnet_b3	IDRiD	baseline	0.116	+0.000
baseline_efficientnet_b3	IDRiD	clahe	0.111	-0.006
baseline_efficientnet_b3	IDRiD	colour_xfer	0.127	+0.011
retfound_full	Messidor-2	baseline	0.395	+0.000
retfound_full	Messidor-2	clahe	0.606	+0.211
retfound_full	Messidor-2	colour_xfer	0.526	+0.131
retfound_full	IDRiD	baseline	0.139	+0.000
retfound_full	IDRiD	clahe	0.293	+0.154
retfound_full	IDRiD	colour_xfer	0.118	-0.021

Inference-only; RETFound was trained at 224 and is run at 384 with a bicubically interpolated positional embedding (off-distribution test).

QWK by model x dataset x resolution

Model	Dataset	res	QWK	acc	macro-F1	referable AUC
baseline_efficientnet_b3	APTOS-internal	224	0.876	0.803	0.644	0.980
baseline_efficientnet_b3	Messidor-2	224	0.499	0.626	0.360	0.842
baseline_efficientnet_b3	IDRiD	224	0.760	0.497	0.441	0.962
baseline_efficientnet_b3	APTOS-internal	384	0.824	0.756	0.535	0.977
baseline_efficientnet_b3	Messidor-2	384	0.425	0.616	0.330	0.866
baseline_efficientnet_b3	IDRiD	384	0.731	0.600	0.435	0.984
retfound_full	APTOS-internal	224	0.882	0.783	0.632	0.967
retfound_full	Messidor-2	224	0.487	0.624	0.368	0.817
retfound_full	IDRiD	224	0.744	0.451	0.419	0.885
retfound_full	APTOS-internal	384	0.886	0.776	0.636	0.975

Model	Dataset	res	QWK	acc	macro-F1	referable AUC
retfound_full	Messidor-2	384	0.553	0.643	0.409	0.851
retfound_full	IDRiD	384	0.652	0.413	0.389	0.856

Internal->external QWK drop, and change vs 224

Model	External	res	QWK drop	delta vs 224
baseline_efficientnet_b3	Messidor-2	224	0.377	+0.000
baseline_efficientnet_b3	IDRiD	224	0.116	+0.000
baseline_efficientnet_b3	Messidor-2	384	0.399	+0.022
baseline_efficientnet_b3	IDRiD	384	0.093	-0.024
retfound_full	Messidor-2	224	0.395	+0.000
retfound_full	IDRiD	224	0.139	+0.000
retfound_full	Messidor-2	384	0.333	-0.062
retfound_full	IDRiD	384	0.234	+0.096

S10.6 Post-hoc temperature scaling

Calibration mode: **held-out:APTOS-val**
Bins: 15 equal-width, top-1 confidence

Fitted temperature (per model)

Model	T	reading
EfficientNet-B3 (baseline)	1.523 (bootstrap 95% CI 1.32-1.71)	over-confident -> softened (T>1)
ResNet-50 (baseline)	1.160 (bootstrap 95% CI 1.01-1.30)	over-confident -> softened (T>1)
RETFound (linear probe)	0.503 (bootstrap 95% CI 0.43-0.58)	under-confident -> sharpened (T<1)
RETFound (full fine-tune)	0.616 (bootstrap 95% CI 0.54-0.70)	under-confident -> sharpened (T<1)

ECE before -> after, by dataset

Model	Dataset	n	ECE raw	ECE scaled	Brier raw	Brier scaled	NLL raw	NLL scaled
EfficientNet-B3 (baseline)	APTOS-internal	733	0.101	0.043	0.300	0.280	0.645	0.549
EfficientNet-B3 (baseline)	Messidor-2	1744	0.279	0.213	0.644	0.594	1.742	1.277
EfficientNet-B3 (baseline)	IDRiD	455	0.294	0.195	0.764	0.692	1.667	1.328
ResNet-50 (baseline)	APTOS-internal	733	0.058	0.053	0.311	0.309	0.608	0.593
ResNet-50 (baseline)	Messidor-2	1744	0.265	0.240	0.650	0.633	1.583	1.420
ResNet-50 (baseline)	IDRiD	455	0.260	0.223	0.778	0.749	1.445	1.357
RETFound (linear probe)	APTOS-internal	733	0.182	0.045	0.395	0.344	0.815	0.753
RETFound (linear probe)	Messidor-2	1744	0.121	0.100	0.594	0.591	1.181	1.190

Model	Dataset	n	ECE raw	ECE scaled	Brier raw	Brier scaled	NLL raw	NLL scaled
RETFound (linear probe)	IDRiD	455	0.127	0.299	0.758	0.874	1.462	1.804
RETFound (full fine-tune)	APTOS-internal	733	0.179	0.081	0.369	0.335	0.748	0.721
RETFound (full fine-tune)	Messidor-2	1744	0.095	0.090	0.563	0.558	1.158	1.179
RETFound (full fine-tune)	IDRiD	455	0.129	0.229	0.715	0.778	1.377	1.634

Temperature scaling is monotonic in the logits, so accuracy and QWK are unchanged (verified: `QWK_raw == QWK_scaled`). Only the probability calibration changes.

S10.7 Feature-space out-of-distribution score versus entropy as an error detector

Model	Dataset	n	errors	AUROC(OOD)	AUROC(entropy)	diff (OOD-entropy), 95% CI	verdict
baseline_efficientnet_b3	APTOS-internal	733	144	0.652	0.841	-0.189 (-0.243,-0.136)	entropy better
baseline_efficientnet_b3	Messidor-2	1744	653	0.472	0.681	-0.210 (-0.244,-0.176)	entropy better
baseline_efficientnet_b3	IDRiD	455	229	0.486	0.592	-0.106 (-0.174,-0.032)	entropy better
baseline_resnet50	APTOS-internal	733	155	0.644	0.788	-0.145 (-0.197,-0.091)	entropy better
baseline_resnet50	Messidor-2	1744	674	0.474	0.652	-0.178 (-0.209,-0.146)	entropy better
baseline_resnet50	IDRiD	455	251	0.506	0.547	-0.041 (-0.106,+0.018)	no sig. difference
retfound_linprobe	APTOS-internal	733	177	0.663	0.695	-0.032 (-0.101,+0.030)	no sig. difference
retfound_linprobe	Messidor-2	1744	716	0.479	0.649	-0.170 (-0.206,-0.135)	entropy better
retfound_linprobe	IDRiD	455	302	0.427	0.648	-0.221 (-0.322,-0.123)	entropy better
retfound_full	APTOS-internal	733	159	0.637	0.516	+0.120 (+0.040,+0.197)	OOD better
retfound_full	Messidor-2	1744	655	0.452	0.633	-0.181 (-0.215,-0.144)	entropy better
retfound_full	IDRiD	455	250	0.556	0.491	+0.065 (-0.026,+0.150)	no sig. difference

APTOS-internal rows are a control: the OOD detector is fit on this same distribution, so a large advantage there (if any) reflects in-domain behaviour, not domain-shift detection.

S10.8 Classifier-head depth and activation sweep (frozen features, 210 heads × 3 test sets)

MCC, Matthews correlation coefficient.

Model(s): efficientnet_b3@aptos_internal, efficientnet_b3@idrid, efficientnet_b3@messidor2, retfound_vit_large_patch16@aptos_internal, retfound_vit_large_patch16@idrid, retfound_vit_large_patch16@messidor2 Grid: depths [0, 3, 4, 5, 6, 7] x activations ['gelu', 'identity', 'relu', 'tanh'] x seeds [0, 1, 2, 3, 4] Width policy: constant Total runs: 630

Seed noise floor

Standard deviation across seed replicates, within each fixed (model, depth, activation) cell. A depth effect smaller than this is not a depth effect.

- median within-cell SD of MCC: 0.0169
- max within-cell SD of MCC: 0.0603

Mean MCC by depth x activation (averaged over seeds)

depth	gelu	identity	relu	tanh
0	nan	0.4135	nan	nan
3	0.414	0.4236	0.4193	0.4261
4	0.422	0.4174	0.4226	0.423
5	0.4266	0.4085	0.4279	0.4154
6	0.4164	0.4028	0.4132	0.4185
7	0.4196	0.4142	0.4097	0.4247

Mean balanced accuracy by depth x activation

depth	gelu	identity	relu	tanh
0	nan	0.4947	nan	nan
3	0.4839	0.4943	0.4839	0.4961
4	0.4868	0.4932	0.4854	0.4924
5	0.4824	0.4847	0.4864	0.4901
6	0.4686	0.486	0.4744	0.4912
7	0.4751	0.487	0.4779	0.4933

Best cell

- model `efficientnet_b3@aptos_internal`, depth 4, activation `gelu` -> mean MCC 0.7009

Treat ‘best’ as provisional until you have checked it against the noise floor above. On a grid this size the top cell is frequently the luckiest cell rather than the best architecture.

Depth effect against the noise floor

- best-minus-worst depth, averaged over activations and seeds: 0.0085
- median within-cell seed SD: 0.0169
- ratio: 0.51x

A ratio below about 2 means depth did not move this task beyond what reseeding moves it on its own.

How to read this

Compare the spread across the depth column against the seed noise floor above. If the best and worst depths differ by less than roughly two within-cell standard deviations, the honest conclusion is that depth did not matter on this task at this width – which is a result worth reporting, not a failed experiment.

Confusion matrices for all 630 cells are in confusion_matrices.json, keyed `model|depth|activation|seed`, as raw integer counts over classes [‘0 none’, ‘1 mild’, ‘2 moderate’, ‘3 severe’, ‘4 proliferative’].